INGB417 FACILITIES DESIGN

GA3 EXAM 2026

Tipe Assessering/ Type of Assessment:	**GA Assignment**
Kwalifikasie/ Qualification:	**B Eng (Industrial)**
Modulekode/ Module code:	**INGB417**
Tydsduur/ Duration:	**5-12 June 2026**
Module beskrywing/ Module description:	**Facilities Design**
Maks/ Max:	**100**
Eksaminator(e)/ Examiner(s):	**Willemien Kruger (BEng Industrial, MEng, PrEng)**
Internal Moderator:	**Carla Troskie (BEng Industrial)**
External Examiner:	**Su-Anrie Aucamp (BEng Industrial, PrEng)**

Instructions:

Read the DESIGN BRIEF as given below, then do appropriate and relevant research to develop your FACILITY DESIGN. This is to be presented in a Facility Design Report (like the Group Assignment report), including all relevant chapters to demonstrate your understanding of the Engineering Design process. An *online* Q&A session will be held on Friday 5 June 13h00-13h30

(Teams link: https://teams.microsoft.com/l/meetup-join/19%3ameeting_MDEwZWVjNTAtZmQ2NC00YmZmLWE2N2YtNzQ1YTIyNGY1NjA2%40thread.v2/0?context=%7b%22Tid%22%3a%22b14d86f1-83ba-4b13-a702-b5c0231b9337%22%2c%22Oid%22%3a%22b9f2f887-cab4-4485-8af9-242477d1508c%22%7d) to clarify expectations and answer any questions you may have.

Your FACILITY DESIGN **Report** must be uploaded (pdf), to eFundi on **Friday 12 June 2025 by 8h00.**

Your MFP must be uploaded as 2 separate documents: the original source file (Draw.io or Rayon format) and a PDF version. You will be penalized if both versions are not uploaded.

You do not have to submit a printed report.

The body of the report should be no longer than 40 pages. Appendices may be included and will not count towards the 40 pages. Appendices will only be considered if they are appropriately referred to in the body of the report. No additional files, drawings or spreadsheets will be considered by the examiners.

Remember that more words do not necessarily reflect better insight and since this subject is about a Layout plan and the supporting key assumptions and calculations, focus on getting that right.

No changes to the report will be accepted once it is uploaded to eFundi. No late or resubmissions will be accepted. The MFP you bring to the interview will be matched with the one submitted.

Exit interviews will be done with an examiner panel on **Wednesday 17 June (8h30-16h00) as well as Thursday 18 June 2026 (9h00-13h00)**. The detail schedule and venue have been communicated in the week of 25 May 2026 for individual planning purposes. Please bring only a colour **PRINTED A2 minimum size**, printed on normal paper **(or larger if you wish) of the MASTER FACILITY PLAN** to the interview to be part of the discussion. Please note: If no Alternatives and/or Final MFP design was presented in the report, GA3 is failed and no interview will be scheduled.

The main purpose of this assignment is to assess your ability to make use of a systematic engineering design approach (as applied to facility design) as the Graduate Attribute (GA) 3 to graduate as an Industrial Engineer. You must obtain at least 50% for the exam report as well as passing GA3. Failure to demonstrate competence in this GA will result in failure of this assessment and the module.

DESIGN BRIEF: **<u>DIGMORE mini shovel</u>**

The DigMore Co. is a shovel manufacturing company with an established history in the industry and a broad customer base. The company's management team decided to expand their product lines with a new shovel design: an all-purpose folding shovel ideal for camping. With the recent surge in market demand for this shovel design, the DigMore Co.'s current production facility does not have sufficient capacity to produce this new product, a new facility is needed dedicated to this. Management has asked for the plans for a new facility to make strategic decisions. You can assume that access to the facility from public roads will be a requirement. The facility must leave about 10% space available for future expansion needs.

DIGMORE requires you, as Industrial Engineer, to *<u>apply a systematic engineering design approach</u>* to develop a comprehensive FACILITIES LAYOUT for the new **manufacturing** facility. Apart from providing space for the main production processes, the facility needs to accept delivery vehicles/trucks from *suppliers* bringing raw materials to offload, it needs to store all materials inside the facility, as well as provide a place for *customers* collecting orders, or their own trucks loading orders for delivery to customers.

THE PRODUCT – mini shovel

The shovel is similar to the design shown in Figure 1 below, with a slightly shorter handle.

Figure 1: Folding Shovel Concept Design The

folding shovel consists of three parts:

- *Back of handle*: 150 mm long section of round tube narrowed at one end with an external thread cut on this end and a knurled grip at the other end.
- *Middle section of handle*: 150 mm long section of round tube with an internal thread cut to connect this section of the handle to the handle grip part. This part is attached to the shovel head using a hinge attached with a rivet.
- *Shovel head*: The shovel head and hinge connector part are stamped from sheet metal, bent to the correct shape and riveted together. The finished shovel head is 150 mm long and 120 mm wide.

PRODUCT AND MANUFACTURING INFORMATION

The easiest way to understand the manufacturing process is to watch the current process where a variety of shovels are made: https://www.youtube.com/watch?v=vnK7-VnF_Zw
[please note that the video is for background and information purposes only]
In the new facility they will be doing things similarly, but they may need to purchase more machines and get more labour to fulfil demand. You are required to help management with these calculations on work and capacity demand for this particular shovel only.

Table 1 provides the main production steps with estimated production times for this shovel. Some of these steps will be visible in the video and where not, please use own discretion and research to make own assumptions.

Table 1: Folding Shovel Production Steps and Estimated Processing Times

Production Step	Estimated Processing Time
Cut handle	15 s per cut (number of pipes per cut depends on chosen equipment)
Internal thread on the middle section	20 s per pipe
Stamp form the end of the grip section to narrow	5 s per pipe
Knurling and external thread on the grip section	1 min per pipe
Stamp hinge connector	2 s per part
Bend hinge connector	5 s per part
Stamp shovel head	2 s per part
Sharpen the shovel head blade	2 min per shovel
Serrate shovel head blade edge	1 min per shovel
Shape/bend the shovel head	2 s per shovel head
Rivet hinge connector to shovel head	5 s per shovel
Heat treat shovel head subassembly	Depends on equipment
Sandblasting shovel heads	Depends on equipment
Hand polish to remove burrs	1 min per shovel head
Powder-coat all parts (black colour)	Depends on equipment
Assemble the shovel head and handle section	30 s per shovel
Packaging (first into small bags that are bought from an outside supplier, then into boxes)	Depends on workstation design

Raw Materials and Suppliers:

The raw materials used to make the shovels are aluminium round tubes and aluminium sheets. The current supplier for the round tube, with an outer diameter of 31.70 mm and a wall thickness of 3.18

mm, is Aluminium Trading (https://www.aluminiumtrading.co.za/aluminium-extrusions/aluminiumround-tubes/). The same supplier also supplies aluminium sheets or coils of rolled aluminium with a thickness of 2.5 mm (https://www.aluminiumtrading.co.za/rolled-aluminium-products/), although the management team will consider a different supplier suggested by the facility designer if preferred.

Facility Requirements:

Some specific requirements management needs are to have the powder coating in a separate totally enclosed area within the building, as the fumes is a safety consideration.

If any other details are needed to determine space requirements, you can make assumptions. You are also required to define your own departments to develop alternative block layouts for management to evaluate. Process flow charts must form part of their Quality Management System. You may use other tools and techniques to design an effective Facility Layout.

Other policies to consider include:

Manufacturing Operating Times

The company operates for one 9-hour shift per day for 5 days a week. Operators receive a 30-minute lunch break and two 15-minute tea breaks on a full shift. Overtime is possible if flexible capacity is required, but management would prefer to avoid it where possible. A total of 220 workdays can be assumed per year.

Inventory Levels

The company's inventory policy stipulates that 2 weeks' worth of raw materials must be kept in stock due to long supplier lead times. Finished goods storage must be sufficient for 4 weeks of stock, as the company has an international market and must ship larger orders at a time. Work-in-Progress inventory levels and space allocation are based on the design of the facility and must be stipulated by the designer.

Deliveries and Collections

The facility must make provision for small order collection on-site by customers and the dispatch of large orders from the facility. As the DigMore makes use of its own fleet of delivery trucks for large orders, space must be allocated on the premises for the truck fleet. The raw material delivery and finished goods dispatch schedules can be negotiated, but all deliveries and collections must occur during working hours on weekdays. The definition of small and large order units must be determined by the facility designer.

The management team also provided the following assumptions and constraints for the new facility:

- The current projected market **demand is 400 000 folding shovels per year**.
- A scrap rate of 5% each can be assumed for tube cutting and sheet stamping processes. No other scrap rates are required.
- Space must be allocated to allow for 10% expansion to meet projected growth within the next 5 years.
- All packaging is performed by hand to ensure a final visual check before shipping.
- Powder-coating and sandblasting are assumed to have a negligible effect on the mass of the final products.

FACILITY DESIGN EXPECTATIONS

The following provides detail expected to see in the Final Design.

- **Materials Receiving and Storage Design considerations**
  - Clear *receiving space* for all suppliers,
  - *storage* of all raw materials and components including details of 3D space requirements based on calculations, material flow and layout and material handling equipment recommendations.
  - *Staging* of materials and components at manufacturing and assembly.
- **Manufacturing Operations Design**
  - *Process flow charts* to indicate clear understanding of the manufacturing processes required, including insight into bottlenecks, workstations and departments.
  - *Production capacity analysis* (including detail calculations) compared to forecast demand to improve on process design so that the manufacturing processes can meet demand. Some changes will have to be made to meet future demand.
  - *Capacity analysis for all manufacturing processes* including machine and personnel requirements organized in workstations.
  - *Recommended machinery* for all critical manufacturing processes.
  - Operator and material movement requirements including movement between manufacturing processes and movement of material from raw material storage to the WIP in manufacturing areas and then to finished goods storage. This can be summarised as flow and should be indicated on the final layout.
  - Detailed workstation layout and space planning for all workstations and WIP stations where required.
- **Finished Product Storage and Shipping Operations Design**
  - Finished product and order *storage capacity* including details of 3D space requirements based on calculations, flow and layout and equipment and materials handling recommendations.
  - Clear and logical *staging* and *dispatch* operation space for customers getting their small orders or
  - Shipping operations where the company has their own trucks taking large orders to distribution centres.

- **Supporting Facilities & Services Design**
  - Overall security considerations, entrance, exit and goods vehicle flow on-site including parking.
  - Parking facilities for all employees on site. Personnel include the facility's management, administrative staff, security personnel and operators. Reasoning and calculations for determining parking space must be provided.
  - Restrooms, lockers and cafeteria access must be available to all personnel.
  - Office space must be allocated to the Senior Plant Manager, Operations Manager, Inventory Manager, the administrative team, including Reception, Purchasing, Sales, Finance, Marketing and Human Resources employees, and the plant supervisors. The number of supervisors will depend on the facility design and must be clarified by the design team.
  - Waste handling and management.
  - Utility requirements including water, electricity, plumbing etc.

THE DELIVERABLES

This project requires you to apply a systematic engineering design approach to develop a **comprehensive** facilities layout (Master Facility Plan) that identifies and meets the needs of the client.

DIGMORE has specifically asked you to include the following in your report:

1. A set of detailed facility design requirements.
2. A thorough capacity/demand analysis to clearly demonstrate production capacity needs.
3. At least three alternative design solutions. All considerations must be communicated.
4. A detailed justification of chosen design(s) including evaluation criteria.
5. A detailed master facilities plan that includes sufficient details of workstation, material handling, storage, receiving, dispatch and personnel space requirements to present to management.
6. A well-prepared Facility Design Report.

The report must include comprehensive evidence to indicate that you followed a systematic Engineering Design process and summarise your key assumptions and capacity calculations.

An opportunity to explain how you approached this project will be given in an EXIT INTERVIEW where you will have 10 minutes to explain how you approached the design problem, what you have done practically to develop design options, and explain the logic of your Master Facility Plan. This is not a presentation, but question and answers session with examiners to test insight. Remember to bring with the 2D layout colour printed in A2 format (min size) to use as reference.

Attached the Rubric that will be used for marking each student individually.

Facilities Design – INGB417 – Examiner & Moderator mark sheet for use in 2026		
Student details		
Initials	**Surname**	**Student number**
Examiner or Moderator details (circle)		
Initials	**Surname**	**Internal or External (circle)**

GA 3: Engineering Design
Design creative solutions for complex engineering problems and design ***systems***, ***components*** or ***processes*** to meet identified needs.

EVALUATION CRITERIA			
	Weighting	**Mark (Interview)**	**Mark (Report)**
Clear interpretation of the design problem and ***formulation/calculation of design requirements or specifications.*** (includes appropriate use of ***analytical tools*** and techniques.)	***/30***		
Practical ***insight shown*** during the design process and demonstrate the **/20** justify assumptions and decisions made in the design solution	ability to		
Defendable final design solution to meet the design requirements **/30** (Receiving & Dispatch, Production processes, Storage, Warehouses, Offices, Support functions all in MFP)			
Effective **application** of a structured and rigorous ***design process,*** **/10** including comparing alternatives systematically and selecting a final design based on defensible criteria			
Effective ***communication*** of the design process and solution	***/10***		
FINAL MARK (Interview 50: Report 50)	***/100***	**Interview mark**	**Report total**
Moderators / examiners comments and marks			
	FINAL MARK		

ASSESSMENT OF GA 3 (ENGINEERING DESIGN)	
GA Achieved? YES/NO	

Comments:			
Assessment Date:		Signature:	